Experiment 3

Title

Write a Program to Solve the Water Jug Problem

Aim

To write a Python program to solve the Water Jug Problem using state-space search.

Objective

To understand state-space search by obtaining the required quantity of water using two jugs of different capacities.

Algorithm

Step 1: Initialize two jugs with capacities 4 L and 3 L.

Step 2: Fill the 4 L jug.

Step 3: Pour water from the 4 L jug into the 3 L jug.

Step 4: Empty the 3 L jug.

Step 5: Pour the remaining water from the 4 L jug into the 3 L jug.

Step 6: Fill the 4 L jug again.

Step 7: Pour water into the 3 L jug until it is full.

Step 8: The remaining water in the 4 L jug is the required quantity (2 L).

Flowchart

Start

|

▼

Initialize Jugs

|

▼

Fill 4L Jug

|

▼

```
Pour into 3L Jug
        |
        ▼
Empty 3L Jug
        |
        ▼
Repeat Steps
        |
        ▼
Required Water?
     |        |
    No       Yes
     |        |
     └────────▶ Display Result
             |
             ▼
           Stop
```

Python Program

```
jug1 = 4
jug2 = 3

x = 0
y = 0

steps = [
   (4, 0),
   (1, 3),
   (1, 0),
   (0, 1),
   (4, 1),
```

```
    (2, 3)
]

print("Steps to get 2 Litres:\n")

for i, (x, y) in enumerate(steps, 1):
    print(f"Step {i}: ({x}, {y})")

print("\n2 Litres obtained in Jug1.")
```

Sample Output

Steps to get 2 Litres:

Step 1: (4, 0)

Step 2: (1, 3)

Step 3: (1, 0)

Step 4: (0, 1)

Step 5: (4, 1)

Step 6: (2, 3)

2 Litres obtained in Jug1.

Result

Thus, the Water Jug Problem was solved successfully, and the required 2 litres of water was obtained.

Conclusion

The Water Jug Problem demonstrates how state-space search can be used to solve real-world problems by applying a sequence of valid operations.